

GMF Investments - Portfolio Optimization Report

Model Performance Summary:

ARIMA: MAE=74.80, RMSE=81.18, MAPE=32.7%

LSTM: MAE=72.75, RMSE=78.98, MAPE=31.8%

Selected Model: LSTM

Portfolio Recommendations:

Max Sharpe Portfolio (Growth):

TSLA: 64.8%

BND: 0.0%

SPY: 35.2%

Minimum Volatility Portfolio (Conservative):

TSLA: 5.3%

BND: 85.6%

SPY: 9.1%

Backtest Performance (Aug 2024 - Jul 2025):

Strategy Return: 35.2% vs Benchmark: 11.7%

Strategy Sharpe: 0.94 vs Benchmark: 1.02

Risk Assessment:

Strategy Volatility: 51.9%

Benchmark Volatility: 12.4%